DATA VISUALIZATION PROJECT · AY 2025–2026

Does AI Boost Grades While Undermining Real Understanding?

The Paradox of Artificial Performance

DELIVERABLE 2 — TECHNICAL REPORT WITH DATASET DESCRIPTION

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Abstract

This technical report presents the three datasets underpinning the data visualization project "Does AI Boost Grades While Undermining Real Understanding?" The central research question investigates whether heavier ChatGPT use is associated with a growing asymmetry between perceived grade improvement and perceived critical thinking development.

1. Dataset Origins and Structure

1.1 Dataset 1 — Aristovnik et al. (2024): Global ChatGPT Student Survey

Source and citation: Aristovnik, A., et al. (2024). *Students' Perception of ChatGPT: A Global Survey.* Mendeley Data, DOI: 10.17632/ymg9nsn6kn. Published in PLOS ONE, 2025. Available via CovidSocLab (https://www.covidsoclab.org/chatgpt-student-survey/).

Collection method: Online questionnaire distributed through an international network of over 200 academic partners. Data collected from higher education students aged 18+ across 109 countries. Participation was voluntary and anonymous. The survey comprised 11 sections and approximately 42 questions.

The dataset contains 23,218 individual responses and 174 variables (columns A–FR in the Excel file). Variables cover: sociodemographic characteristics (nationality, gender, age, field of study, study level), ChatGPT usage patterns (adoption, intensity, tasks), capabilities perceptions, ethical concerns, satisfaction, study outcomes, skills development, labour market views, emotional responses, and personal study information.

Variable	Question	Type	Scale / Values
Q13	Have you ever used ChatGPT?	Binary	1=Yes, 2=No
Q15	Usage intensity	Ordinal	1=Rarely … 5=Extensively
Q10	Field of study	Nominal	1=Arts, 2=Social, 3=Applied, 4=Natural
Q27b	ChatGPT can improve my grades	Likert	1=Strongly disagree … 5=Strongly agree

Variable	Question	Type	Scale / Values
Q29e	ChatGPT can improve my critical thinking	Likert	1=Strongly disagree … 5=Strongly agree
Q22j	ChatGPT hinders learning (does work for students)	Likert	1=Strongly disagree … 5=Strongly agree
Q32l	Emotion: Curious (while using ChatGPT)	Frequency	1=Never … 5=Always
Q4	Country of study	Nominal	109 unique values

Table 1.1 — Key variables in Dataset 1 (Aristovnik et al., 2024). Full variable list spans 174 columns.

1.2 Dataset 2 — Survey_AI (Faculty of Cybernetics)

This dataset consists of 91 responses collected at the Faculty of Cybernetics via an internal survey on students' perceptions of Artificial Intelligence. The questionnaire covers AI knowledge level, information sources, perceived societal impacts (dehumanization, job replacement, problem solving, societal control), economic concerns, personal feelings towards AI, perceived utility (scored 1–10), advantages and disadvantages for education, and student demographics (gender, year, major, GPA, passed exams).

Variable	Description	Type	Key Statistics
Q1	Self-rated AI knowledge (1–10)	Continuous	Mean=5.91, SD=1.97
Q5	Dominant feeling towards AI	Nominal	1=Curiosity (68.1%), 2=Fear (13.2%)
Q7	Perceived utility (1–10)	Continuous	Mean=7.44, SD=2.16, Median=8.0
Q8–Q11	Advantages/disadvantages (1–3 scale)	Ordinal	1=agree, 3=disagree
Q13	Year of study	Ordinal	1=1st year (37.4%), 2=2nd year (62.6%)
Q16	GPA	Continuous	Mean=7.80, SD=0.96

Table 1.2 — Key variables in Dataset 2 (Survey_AI, n=91).

Dataset 2 uses a different scale from Dataset 1: Q8–Q11 are coded 1=agree, 2=neutral, 3=disagree (3-point), not the standard 5-point Likert. All analyses and visualizations account for this difference. The two datasets are not mergeable at the individual level.

1.3 Dataset 3 — students_ai_usage (Semi-Synthetic)

This dataset contains 100 student records with 9 variables: age (14–19 years), education level (school/college), daily study hours, AI usage status (Yes/No: 40 AI users, 60 non-users), AI tool used, purpose of AI use (Research/Homework/Coding), grades before and after AI adoption, and daily screen time. It provides a simplified longitudinal structure suited for before/after grade visualization.

Variable	Description	Type	Range / Values
uses_ai	Whether student uses AI tools	Binary	Yes (n=40), No (n=60)

Variable	Description	Type	Range / Values
purpose_of_ai	Primary purpose of AI use	Nominal	Research (14), Coding (13), Homework (13)
grades_before_ai	Grade score before AI adoption	Continuous	Range: 55–75 (non-users); 60–72 (users)
grades_after_ai	Grade score after AI adoption	Continuous	Range: varies; AI users mean=75.15
education_level	School or college	Binary	School (n=52), College (n=48)
age	Student age in years	Continuous	Mean=16.49, SD=1.70, Range: 14–19

Table 1.3 — Variables in Dataset 3 (students_ai_usage, n=100).

CRITICAL SYNTHETIC CAVEAT: Dataset 3 is semi-synthetic. The grade delta for non-users is exactly 0.00 (grades_before = grades_after for all 60 non-users) — an artifact of simulation. This is statistically unrealistic. The t-test comparing AI vs. non-AI users (t=21.49, p≈0) is trivially significant because of this artifact. All visualizations derived from this dataset are explicitly labeled as illustrative and do not support causal inference.

2. Exploratory Data Analysis (EDA)

2.1 Dataset 1 — ChatGPT Adoption and Usage Intensity

Of 23,218 students, **16,010 (69.0%)** have used ChatGPT (Q13=1) and 6,404 (27.6%) have not (Q13=2). The remaining 804 responses (3.5%) have a missing or invalid Q13 value. Among the 16,010 users, usage intensity (Q15) is concentrated in the low-to-moderate range: 30.1% use ChatGPT Occasionally, 28.4% Moderately, and only **4.3% Extensively** — this last subgroup is analytically critical as it produces the strongest paradoxical signal (see Section 2.2).

By field of study, Applied Sciences (74.4%) and Natural & Life Sciences (68.1%) show the highest adoption rates, while Arts & Humanities lags at 66.5%. Social Sciences (67.3%) sits just below the overall average. These differentials are consistent with the literature: STEM and quantitative fields tend to adopt computational tools faster.

Usage Intensity	Code	n (valid)	% of users	Analytical role
Rarely	1	3,375	21.2%	Baseline / control group
Occasionally	2	5,100	32.0%	Majority group
Moderately	3	4,581	28.8%	Regular use
Considerably	4	2,175	13.7%	Heavy use
Extensively	5	694	4.3%	Key paradox signal group
Missing / NA	—	7,291	—	Excluded from intensity analysis

Table 2.1 — Usage intensity distribution (Dataset 1, n=15,925 valid Q15 responses). Note: Q15 is only asked to ChatGPT users (Q13=1), hence 7,291 missing = 6,404 non-users + 887 users with missing Q15.

2.2 Dataset 1 — The Paradox: Grades vs. Critical Thinking

The central research question of this project concerns whether heavier ChatGPT use is associated with a growing asymmetry between perceived grade improvement (Q27b) and perceived critical thinking development (Q29e). Computing the mean of each variable per intensity level reveals a monotonically increasing gap (Q27b − Q29e):

Intensity	Q27b (Grades)	Q29e (Crit. Think.)	Gap (Q27b−Q29e)	Interpretation
Rarely (1)	2.901	2.709	0.192	Near-symmetric benefit
Occasionally (2)	3.235	2.965	0.270	Gap growing
Moderately (3)	3.466	3.139	0.327	Moderate asymmetry
Considerably (4)	3.761	3.329	0.432	Substantial gap
Extensively (5)	4.082	3.538	0.543	Maximum asymmetry

Table 2.2 — The paradox gap (Q27b − Q29e) by usage intensity. Gap increases by +0.351 points from rare to extensive use. Values computed directly from final_dataset.xlsx.

INTERPRETIVE CAUTION: Q27b and Q29e are self-reported perceptions, not objective outcome measures. The observed gap may partially reflect confirmation bias — heavy users have stronger incentives to perceive grade benefits. The data cannot distinguish between "ChatGPT truly helps grades more than critical thinking" and "heavy users overestimate benefits". Both interpretations are consistent with the observed values.

2.3 Dataset 1 — Dependency and Rationalisation (Q22j)

A counter-intuitive finding emerges from Q22j ("ChatGPT might hinder learning by doing the work for students"): agreement with this concern decreases as usage intensity increases. Rare users show the highest concern (mean=3.352, leaning toward agreement with caution), while extensive users show the lowest (mean=2.834).

This pattern, labeled *"cognitive rationalisation"* in the proposal, has two competing explanations: (1) experienced users genuinely learn to use ChatGPT more effectively and do not perceive learning obstacles; (2) cognitive dissonance between heavy use and potential harm is resolved by lowering perceived concern. Cross-sectional data cannot distinguish between these mechanisms. The 4.3% "Extensively" group (~694 students) is also a numerically small tail and may not be representative of the broader population of intensive AI users.

2.4 Dataset 1 — Emotional Profile

Emotion	Mean (1–5)	Rank	Interpretation
Curious	3.430	1st	Dominant positive orientation
Calm	3.265	2nd	Low-anxiety engagement
Happy	3.123	3rd	Generally positive affect
Hopeful	3.088	4th	Forward-looking attitude
Excited	2.916	5th	Moderate enthusiasm
Bored	2.258	—	Low boredom
Ashamed	1.821	Last-2	Minimal guilt in AI use
Sad	1.731	Last-1	Negligible sadness

Table 2.3 — Emotional profile while using ChatGPT (Q32, Dataset 1, n=16,010 users). Scale: 1=Never, 5=Always.

The emotional profile is predominantly positive and curiosity-driven. Notably, Ashamed (1.821) and Sad (1.731) are among the lowest scores, indicating that students do not experience significant guilt when using ChatGPT — a relevant finding for academic integrity policy discussions. This profile is fully consistent with Dataset 2 findings (see Section 2.5).

2.5 Dataset 2 — Perceptions at the Faculty of Cybernetics

Dataset 2 reveals a paradoxical coexistence in student perceptions: students simultaneously agree that AI has educational advantages and that it represents a disadvantage for the educational process. Mean scores for Q8 (Teaching Advantage), Q9 (Learning Advantage), Q10 (Evaluation Advantage) and Q11 (Educational Process Disadvantage) are all below the midpoint of the 1–3 scale, indicating agreement across all four items.

Variable	Mean	SD	Median	Interpretation
Q7 — Perceived Utility (1–10)	7.44	2.16	8.0	High overall utility perception; polarised
Q8 — Teaching Advantage (1–3)	1.92	0.88	2.0	Strong agreement (below midpoint 2.0)
Q9 — Learning Advantage (1–3)	1.88	0.68	2.0	Strongest agreement (lowest mean)
Q10 — Evaluation Advantage (1–3)	2.25	0.82	2.0	Agreement
Q11 — Educational Disadvantage (1–3)	2.10	1.03	2.0	Agreement — paradox confirmed
Q1 — AI Knowledge Self-Rating (1–10)	5.91	1.97	6.0	Moderate self-assessed knowledge
Q16 — GPA	7.80	0.96	7.9	Above-average academic performance

Table 2.4 — Descriptive statistics for key Survey_AI variables (n=91).

Dominant feelings: Curiosity dominates at 68.1% (n=62), followed by Fear (13.2%, n=12), Indifference (11.0%, n=10), and Trust (7.7%, n=7). By year of study, first-year students show more Fear (17.6% vs. 10.5%) and less Curiosity (61.8% vs. 71.9%) than second-year students, consistent with an experience effect. Main information sources: Internet (81.3%), Social Media (44.0%), Books/Papers (35.2%), Discussions (19.8%); only 6.6% are not informed about AI.

2.6 Dataset 3 — Grade Delta Analysis

Among the 40 AI-using students, the mean grade delta is **+9.83 points** (from 65.33 to 75.15 on a 100-point scale, SD=3.55). By purpose of use, Research yields the highest delta (+10.29 pts), followed by Coding (+10.00) and Homework (+9.15). College students show higher deltas than school students, particularly for Research (college: +12.00 vs. school: +8.57).

The non-user delta of exactly 0.00 is a synthetic artifact that makes the t-test result (t=21.49, p≈0) trivially significant and non-interpretable. This dataset should be treated as illustrative of before/after grade trajectories, not as causal evidence of AI's effect on grades.

3. Data Quality Assessment

3.1 Missing Data Analysis (Dataset 1)

Dataset 1 has structured missing data patterns that are not random but follow the survey design logic. Q13 (ChatGPT usage) has 804 missing values (3.5%) — representing likely incomplete submissions or invalid entries that were not filtered at collection. All downstream variables (Q15, Q27b, Q29e, Q22j, Q32) are only presented to ChatGPT users, creating expected missingness for non-users (6,404 records). The effective analytical population for paradox analysis is 16,010 users.

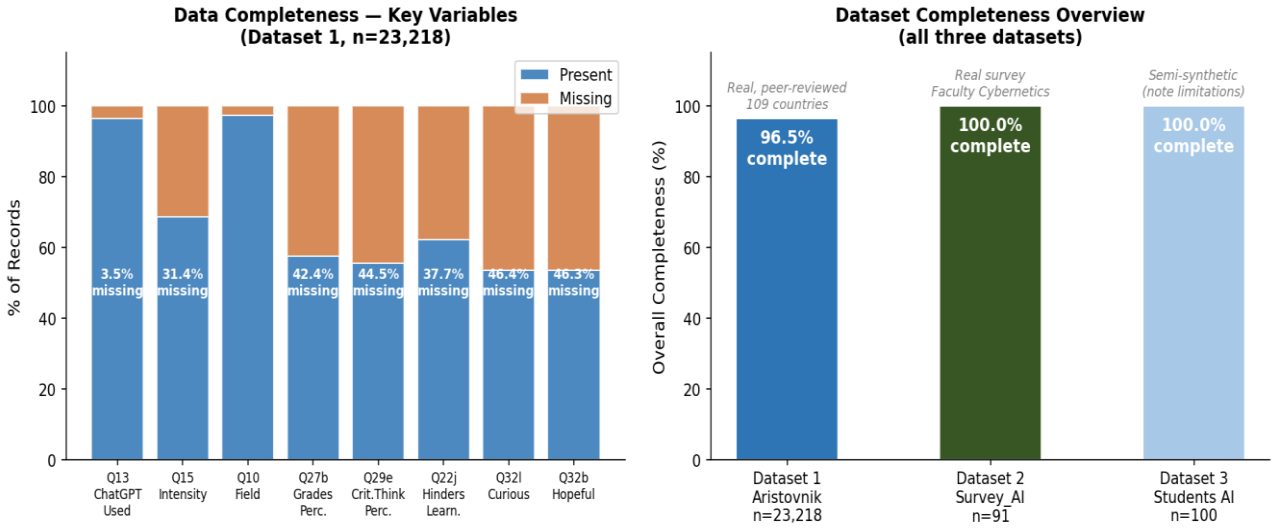

Figure 6 — Left: Data completeness for key variables in Dataset 1. Variables asked only to ChatGPT users (Q27b, Q29e, Q22j) have ~40–45% missing — expected by design, not a quality issue. Right: Overall completeness across all three datasets.

Variable	Missing (n)	Missing (%)	Reason	Impact
Q13	804	3.5%	Incomplete submissions / filter	Low — excluded from analysis
Q15	7,291	31.4%	Design: only asked to Q13=1 users	Expected — 6,404 non-users + 887 skipped
Q27b	9,838	42.4%	Design: only asked to Q13=1 users	Expected — analytical n=13,380
Q29e	10,327	44.5%	Design: only asked to Q13=1 users	Expected — analytical n=12,891
Q22j	8,752	37.7%	Design: only asked to Q13=1 users	Expected — analytical n=14,466
Q10	596	2.6%	Non-response on field of study	Minimal — excluded from field analysis

Table 3.1 — Missing value analysis for key variables in Dataset 1.

3.2 Data Quality: Dataset 2 (Survey_AI)

Dataset 2 has zero missing values across all 35 variables and 91 records — a notable quality strength. No imputation was required. Potential quality concerns are distributional rather than structural: the sample is limited to a single faculty (Cybernetics), creating selection bias. The student population is predominantly technical (Major 2=38 students, Major 1=30, Major 3=23), which may not generalize to other academic contexts. Gender distribution is 59 male (64.8%) and 32 female

(35.2%), reflecting a STEM-skewed sample. GPA range is 5.2–9.7 (mean 7.80), indicating a generally high-performing sample with limited representation of struggling students.

3.3 Data Quality: Dataset 3 (students_ai_usage)

Dataset 3 has no missing values. However, it has significant **synthetic data quality issues** that limit its use for statistical inference. The non-user group shows exactly identical grade distributions before and after (delta=0.00 for all 60 students), which is statistically impossible in real-world data. Age distribution (14–19 years, mean 16.49) is inconsistent with the college education level for some records — a further synthetic artifact. The dataset is appropriate only for *illustrative visualization* of before/after grade trajectories.

3.4 Potential Biases Across Datasets

Dataset	Bias Type	Description	Effect on Visualization
D1	Selection bias	Online survey — may over-represent tech-literate students	May inflate adoption rates
D1	Self-report bias	Perceptions (Q27b, Q29e) not validated against objective outcomes	Gap may reflect confirmation bias
D1	Geographic bias	Strong Asia/Americas representation; European rates may differ	Global figure (69%) may not apply locally
D2	Selection bias	Single faculty (Cybernetics) — technical STEM students only	Perception data may not generalize
D2	Social desirability	Students may under-report fear or over-report curiosity	Emotional profile may be optimistic
D3	Synthetic artifact	Non-user delta=0.00; some age/level inconsistencies	Invalidates statistical tests; illustrative only

Table 3.2 — Potential biases across all three datasets and their expected effects on the visualization.

4. Connection to Visualization Design

4.1 Data-to-Visualization Mapping

The five-tab dashboard structure maps directly onto the datasets and variables described in this report. Each tab addresses a specific hypothesis from the proposal and uses the most appropriate dataset to answer it:

Dashboard Tab	Dataset(s)	Key Variables	Chart Type
Tab 1: Overview	D1	Q13, Q15, Q10	Donut chart, bar chart
Tab 2: The Paradox	D1	Q27b, Q29e, Q15, Q10	Dual-line chart, gap bar chart
Tab 3: Dependency Lens	D1	Q18a–Q18l, Q22j, Q27b, Q29e	Heatmap, 3-metric bar chart
Tab 4: Grade Delta	D3	grades_before/after, purpose, education_level	Slope chart, grouped bar chart
Tab 5: Perceptions	D2 + D1	Q5–Q11, Q32, Q7	Pie chart, bar chart, radar

Table 4.1 — Data-to-visualization mapping across the five dashboard tabs.

4.2 Visual Encoding Decisions Informed by Data Quality

Several data quality findings directly influenced visualization design choices:

- The monotonically increasing paradox gap (Table 2.2) supports the use of a dual-line chart with annotated gap markers — the monotonic trend is visually unambiguous and does not require smoothing.
- The non-user delta=0.00 in Dataset 3 prompted use of a slope chart rather than a bar chart for the grade delta, as slope charts display individual trajectories and make the synthetic artifact visible and explicit.
- The high missingness in Q27b and Q29e (~42–44%) is not a quality problem but a design feature; however, it limits the validity of by-country breakdowns where cell sizes would be too small.
- The single-faculty limitation of Dataset 2 prevents generalization, so all D2 visualizations are explicitly labeled with sample details (n=91, Faculty of Cybernetics).
- The small "Extensively" group (n=694, 4.3%) in Dataset 1 is flagged in all intensity-level charts to prevent overinterpretation of the tail.

4.3 Cross-Dataset Consistency Verification

The three datasets, despite being non-mergeable at the individual level, converge consistently on the project's central narrative:

Hypothesis	Dataset	Result	Status
Heavy ChatGPT use → larger grades/CT gap	D1 (n=16,010)	Gap: +0.192 → +0.543	✓ CONFIRMED
Simultaneous agreement on advantages + disadvantages	D2 (n=91)	Q8–Q11 all < 3.0	✓ CONFIRMED
AI use → measurable grade improvement	D3 (n=40 AI users)	Delta = +9.83 pts (synthetic caveat)	ILLUSTRATIVE

Hypothesis	Dataset	Result	Status
Heavy users less concerned about learning hindrance	D1 (n=16,010)	Q22j: 3.352 → 2.834	✓ CONFIRMED (interpret cautiously)
Curiosity dominant emotion; Trust low	D2 + D1	68.1% (D2) / 3.43 mean (D1)	✓ CONFIRMED (cross-dataset coherence)
Research use → higher grade delta vs. Homework	D3 (n=40)	Research: +10.29 vs. Homework: +9.15	ILLUSTRATIVE (small n)

Table 4.2 — Cross-dataset hypothesis verification. ✓= confirmed on real data; illustrative on semi-synthetic data.

5. Conclusions and Limitations

This report has documented three datasets at different levels of quality and generalizability. Dataset 1 (Aristovnik et al., 2024) is the empirical backbone of the project: real, peer-reviewed, large-scale (n=23,218), and globally representative. It provides robust evidence for the central paradox — heavier ChatGPT use is associated with growing asymmetry between perceived grade improvement and perceived critical thinking gain (gap: +0.192 at rare use → +0.543 at extensive use). Dataset 2 (Survey_AI, n=91) is real but limited in scope to a single technical faculty; it independently replicates the paradox at the subjective level. Dataset 3 (semi-synthetic) is appropriate only for before/after grade illustration and explicitly flagged as such in all visualizations.

Key data quality findings that inform visualization design: (1) missing data in Dataset 1 is structured and expected by survey design, not random; (2) Dataset 2 has zero missing values but carries selection bias; (3) Dataset 3's synthetic artifacts preclude statistical inference. All three limitations have been explicitly incorporated into the dashboard design through annotations, caveats, and contextual labels.

The data quality assessment confirms that the visualization project can deliver a credible and well-calibrated message: AI's effect on learning is heterogeneous and depends critically on how — and how intensively — it is used. The datasets are sufficient to support the proposed narrative with appropriate epistemic humility.